NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503

Autonomous Institution, Affiliated to Anna University – Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CS61C –Business Process Management

CO3: Analyze the process discovery techniques and modeling methods

CO4: Apply the heuristic techniques and product-based design principles to redesign business processes, using real-world case studies such as healthcare systems

1. Examine a **process model for a hospital patient discharge procedure** and describe how **syntactic verification, semantic validation, and pragmatic certification** can be applied to assess its quality, with suitable examples.

2. In a **driving license renewal process at a government office**, applicants frequently follow up due to long approval delays.
Use the **5 Whys technique** and a **Fishbone diagram (People, Procedures, Technology, Policy, Infrastructure, External Dependencies)** to identify root causes and suggest solutions to reduce delays and follow-ups.

3. Consider the Build **IT equipment rental process** consisting of the following steps: *Fill request, Send request to clerk, Open and read request, Select suitable equipment, Check equipment availability, Record recommended equipment and supplier, Forward request to works engineer, Open and examine request, Communicate issues, and Create Purchase Order.*

Classify each step based on the **performer perspective** as **Value-Added (VA), Business Value-Added (BVA), or Non-Value-Added (NVA)** activities.

4.An e-commerce service provider is experiencing significant delays in order fulfillment, leading to increased customer dissatisfaction and complaints. To address this, evaluate key process performance metrics—including cycle time, throughput time, waiting time, utilization, and cost efficiency—to diagnose inefficiencies. Additionally, apply flow analysis techniques to pinpoint bottlenecks and non-value-adding steps in the workflow, and outline strategies for their removal to streamline operations and enhance overall efficiency.

5.Evaluate how heuristic process redesign techniques were applied in a health care institution case study, particularly focusing on the Periodic Meetings and File Handling sub-processes. Analyze the strengths and limitations of the applied heuristics in balancing the Devil's Quadrangle dimensions (time, cost, quality, and flexibility). Further, propose an enhanced redesign strategy by incorporating additional heuristics such as technology adoption or external environment considerations that were not emphasized in the case and the expected impact on overall process performance.

6.Analyze the bottlenecks in the healthcare institution's medical file process using **quantitative process analysis techniques** such as **performance measures, flow analysis, queue analysis, and simulation**. Explain how **heuristic process redesign** addressed issues in **file requesting, posting delays, and periodic meetings**, and discuss the resulting improvements and trade-offs in the **Devil's Quadrangle**

7. A medical office faces delays in requesting patient files due to excessive decision-making, meetings, and fragmented communication. Analyze the **as-is process**, explain how **Contact Reduction, Exception, and Resequencing heuristics** improve it, and describe the **to-be process** where the secretarial office requests files directly after the doctor's contact. Evaluate improvements in **cycle time, coordination, and efficiency**, and relate the **Exception heuristic to the Triage heuristic**.